

Marketing Insight Technical Report (Productdata)

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Project: Product Data Optimization for Digital Marketing

Tools Used: Python (Pandas, NumPy, Regex)

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1. Introduction

This technical report presents the first stage of a broader marketing analytics project aimed at enhancing the quality and usability of raw product data. Specifically, the focus was on preparing data for search engine optimization (SEO), improving product visibility across digital channels, and laying a solid foundation for downstream marketing analysis.

2. Objectives

The main objectives were:

- **Data Cleaning:** Eliminate inconsistencies, handle missing or duplicate data, and standardize naming conventions.
- **Feature Engineering:** Create a concise, SEO-optimized short_title feature to improve product discoverability on digital platforms.

These steps are essential for ensuring accurate marketing performance metrics, enriching product content, and aligning with digital advertising requirements.

3. Data and Methodology

3.1 Data Sources

- **Primary Dataset:** Raw product listing file (Excel format)
- **Total Records:** 3,847 rows
- **Columns:** 6 product-specific attributes including IDs, titles, and descriptions

3.2 Tools and Technologies

- **Python Libraries:**
 - Pandas for data manipulation
 - NumPy for handling null values
 - re (regex) for text pattern recognition and cleaning

3.3 Analytical Workflow

1. Import raw Excel data
2. Inspect missing values and duplicates
3. Normalize column names
4. Drop rows with missing essential fields
5. Create short_title using cleaned and trimmed product titles
6. Export final cleaned dataset for downstream use

4. Results

4.1 Data Cleaning Summary

Action	Outcome
Rows in raw data	3,847
Duplicates removed	217
Rows after cleaning	3,541
Column format standardized	Yes (snake_case applied)
Essential nulls handled	Yes (289 rows removed)
Description fields retained	Yes (optional for later enrichment)

4.2 Missing Value Handling

Column	Missing Values	Action Taken
producttypeid	178	Rows dropped
productlength	178	Rows dropped
bullet_points	1,591	Retained for future use
description	2,144	Retained for enrichment

4.3 Duplicate Removal

- Used `df.duplicated()` to identify and drop 217 exact row matches.
- Resulting dataset has unique product records to avoid skewing analysis.

4.4 Column Standardization

All columns renamed using str.lower().str.replace(' ', '_') convention for consistency:

Before	After
ProductLength	productlength
PRODUCTID	productid
TITLE	title

4.5 SEO Title Feature Engineering

Goal: Create short_title column optimized for platforms with character restrictions and SEO requirements.

Method:

- Removed filler words: “set of”, “includes”, “featuring”, “with”
- Preserved: brand, gender, type, color, size
- Applied regex cleaning for punctuation and spacing
- Character limit: ~50 characters

Examples:

Original Title	SEO-Optimized short_title
ArtzFolio Tulip Flowers Blackout Curtain for Door, Window & Room Eyelets Set of 2 PCS	ArtzFolio Tulip Flowers Blackout Curtain Door...
Marks & Spencer Girls' Pyjama Sets T86_2561C_Navy Mix_9-10Y	Marks & Spencer Girls' Pyjama Sets T862561CNav...
PRIKNIK Horn Red Electric Air Horn Compressor 12V for Cars & Bikes	PRIKNIK Horn Red Electric Air Horn Compressor...

5. Discussion

5.1 Data Quality Impact on Marketing Analytics

Cleaned and standardized data ensures:

- Accurate product categorization
- Reliable A/B testing for product titles
- Consistent metrics across reporting dashboards

5.2 Importance of Title Optimization

Well-structured product titles:

- Increase CTR in search and social platforms
- Improve keyword match relevance in paid ads
- Support better indexing by crawlers and feeds

6. Recommendations

6.1 Future Enrichment Opportunities

- Use NLP or AI models to auto-generate bullet points and short descriptions
- Integrate with product taxonomy APIs to improve categorization

6.2 Scaling for Automation

- Convert title optimization into a reusable function or pipeline
- Apply to future product uploads via a scheduled batch job or cloud function

7. Deliverables

File Name	Description
cleaned_product_data_with_short_title.csv	Final dataset after cleaning & enhancement
Technical Report (this document)	Documentation of methodology and outcomes

8. Conclusion

This report documents a complete and reproducible process for transforming messy product data into a structured, SEO-ready format. The addition of the short_title feature significantly improves usability across marketing platforms and sets the stage for more insightful analysis in the next project phase.

9. References

- Python Documentation: Pandas & Regex
- <https://docs.google.com/spreadsheets/d/1z3Io4xtC2FMif2lskLbDDcRe5OTjoi4l/edit?gid=822295292#gid=822295292>
- <https://colab.research.google.com/drive/1stoiwbUIO6J5F2asiSRrx1XB-2K87l2b?usp=sharing>